

Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency.

Code:

```
geya@LenovoL480: ~/O5/Pra x + v
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <time.h>

int main() {
    int pipefd[2];
    int data[5] = {10, 20, 30, 40, 50};

    pid_t pid;

    struct timespec start, end;

    pipe(pipefd);

    clock_gettime(CLOCK_MONOTONIC, &start);

    pid = fork();

    if (pid == 0) {
        // Child = Consumer

        int received[5];

        close(pipefd[1]);
```

```

        read(pipefd[0], received, sizeof(received));
        printf("Child (Consumer) received:\n");

        for (int i = 0; i < 5; i++) {
            printf("%d ", received[i]);
        }

        printf("\n");
        close(pipefd[0]);
    }

    else {
        // Parent = Producer

        close(pipefd[0]);

        printf("Parent (Producer) sending:\n");

        for (int i = 0; i < 5; i++) {
            printf("%d ", data[i]);
        }

        printf("\n");

```

```

        write(pipefd[1], data, sizeof(data));
        close(pipefd[1]);

        wait(NULL);

        clock_gettime(CLOCK_MONOTONIC, &end);

        double time_taken =
            (end.tv_sec - start.tv_sec) * 1000000.0 +
            (end.tv_nsec - start.tv_nsec) / 1000.0;

        printf("Communication time: %.2f microseconds\n", time_taken);
    }

    return 0;
}

```

Output:

```
geya@LenovoL480:~/OS/Practical/Week 5$ gcc task1.c -o task1
geya@LenovoL480:~/OS/Practical/Week 5$ ./task1
Parent (Producer) sending:
10 20 30 40 50
Child (Consumer) received:
10 20 30 40 50
Communication time: 891.11 microseconds
geya@LenovoL480:~/OS/Practical/Week 5$ |
```